POWER ELECTRONICS AND DRIVES

LABORATORY MANUAL



DEPARTMENT OF ELECTRICAL ENGINEERING

COLLEGE OF ENGINEERING TRIVANDRUM

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Chapter 1

1. Static VI characteristics of Power Device-SCR

1.1 Objective

Aim: To draw the static VI characteristics of an SCR. Find the key parameters like latching current, holding current, resistance etc. Simulate the VI characteristics using any suitable simulation software and compare the results with datasheet values.

1.2 Theory

A Silicon Controlled Rectifier (SCR) is the oldest and the most widely used member of the thyristor family. SCR is a four-layer PNPN device with three junctions. It has three terminals: Anode, Cathode and Gate. The SCR is used as a semi controlled switch. High reliability, high current and voltage ratings are some of the major features which make the SCR suitable for power control in many high-power applications, especially the ones that require phase-control of line-frequency AC power.
The static V-I characteristics of an SCR are shown in Fig. 1.2. The SCR has three basic modes of operation: forward conduction mode(on-state), forward blocking mode (off state) and reverse blocking mode.

1.2.1 Forward Conduction Mode

When the anode is made positive with respect to the cathode, junctions J1 and J3 are forward biased and junction J2 is reverse biased. The reverse biased junction J2 limits the anode current to a few milli amperes (forward leakage current). The SCR is then said to be in the forward blocking mode.

The SCR can be triggered into conduction by applying a positive gate pulse. At higher gate trigger currents, the turn-on can happen at lower anode-to-cathode voltage. If the gate

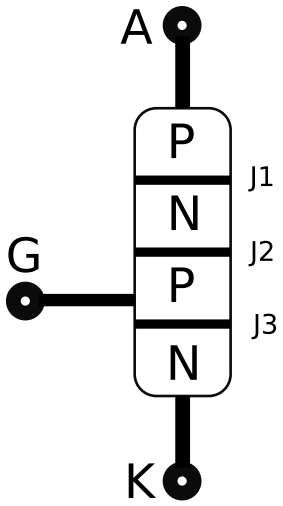


Figure 1.1: Structure of an SCR

current is sufficiently large, the device can be turned on with a small anode to cathode voltage. For lower gate currents, the turn-on happens at higher anode to cathode voltages. The minimum gate current at which a forward-biased SCR turns on is an SCR specification, denoted usually as $I_{gt,min}$.

1.2.2 Forward Blocking Mode

When the anode is made positive with respect to the cathode, junctions J1 and J3 are forward biased and junction J2 is reverse biased. The reverse biased junction J2 limits the anode current to a few milli amperes (forward leakage current). The SCR is then said to be in the forward blocking mode.

When the SCR is in the forward blocking mode, if the anode to cathode voltage is increased to the forward break over voltage, junction J2 breaks down so that the SCR becomes equivalent to a conducting diode. The voltage across the device falls to a small value and the anode current is limited by the load impedance.

1.2.3 Reverse Blocking Mode:

When the cathode is positive with respect to the anode, junctions J1 and J3 (See Fig. 1.1) are reverse biased and J2 is forward biased. Only a small reverse leakage current of the order of a few milliamperes flows and the device is in the reverse blocking mode. When the voltage is increased to the reverse breakdown voltage, the depletion layers at the junctions J1 and J3 break down and the current through the device increases to a high value. The high current at high voltage causes more losses in the device, and the junction temperature may exceed the permissible value resulting in the damage of the device.

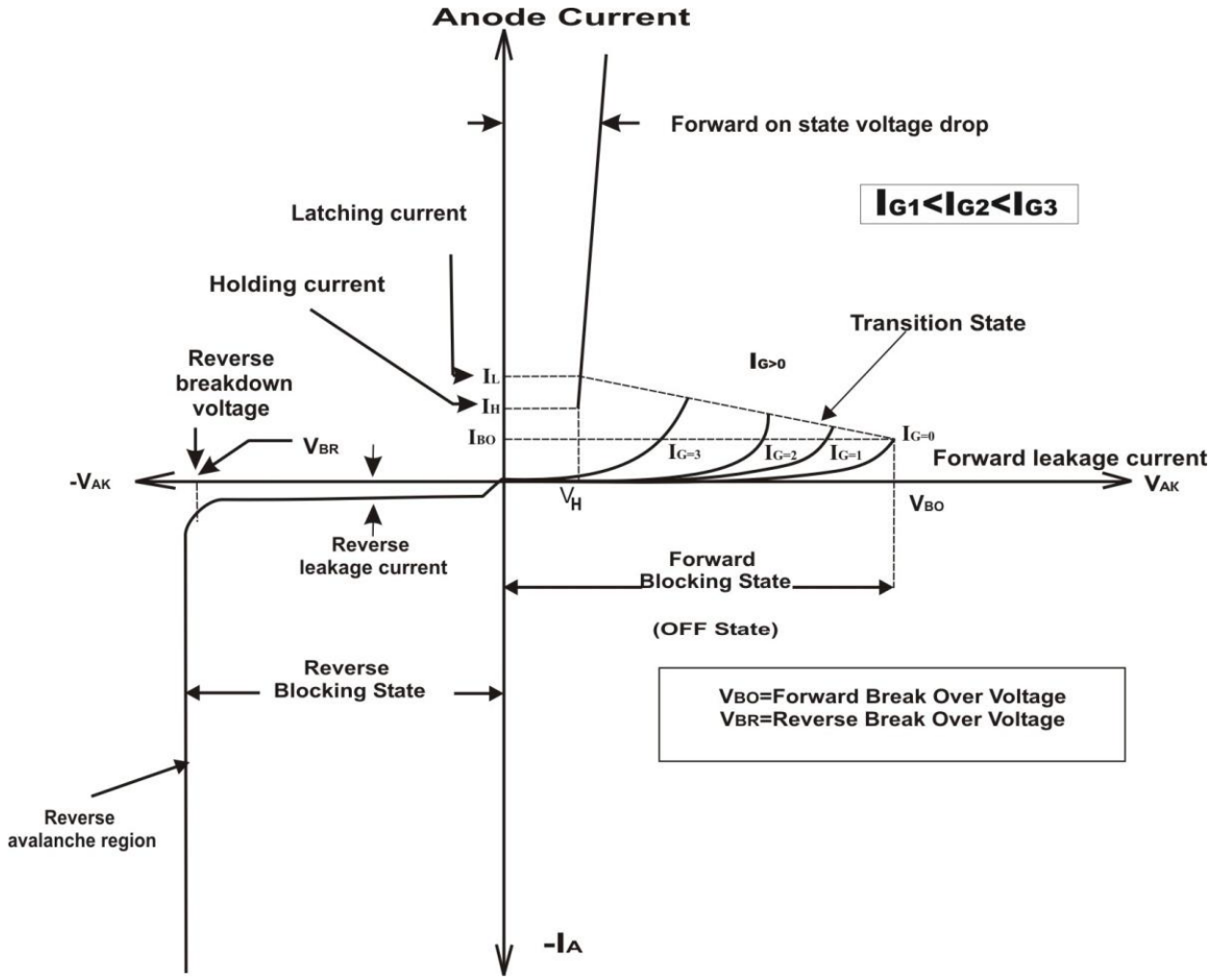


Figure 1.2: V-I Characteristics of SCR

1.3 Design

The design of this circuit is done by selecting TYN616 (SCR). Some essential parameters of the device TYN616 are provided below (for more details, refer the Data sheet. Also redesign the test circuit for other SCRs)

Gate Threshold Voltage,$V_{GT} = 0.8V$
Gate Threshold Current,$I_{GT} = 2mA$
Max. Anode Current,$I_{Amax} = 2A$

$$R_G >= \frac{V_{GG} - V_{GT}}{I_{GT}} \tag{1.1}$$

$$R_{Lmin} = \frac{V_{SS}}{I_{Amax}} \tag{1.2}$$

1.4 Procedure

(Though the procedure will be the same, apply KVL to select the resistances, meters and supply voltages accordingly for SCRs of other ratings)

1.4.1 Determination of I_{gtmin}

The supply voltage is set to 24 V for power circuit and the gate drive circuit supply (5V supply) is kept at minimum voltage position. A fixed resistance (rheostat) of 18 Ohm is connected in the circuit to represent a load. The variable resistance (1000 Ohm rheoststst) is included in the anode circuit to achieve thyristor current variation and is kept in the minimum position. The connections are made as in Figure. The power circuit supply(24V) is switched on. The gate voltage supply is gradually increased till the SCR is turned on and conducts (indicated by a sudden increase in anode current and fall of voltage across the SCR). The gate current at which the SCR just turns on is the value of I_{gtmin}.

1.4.2 Forward conduction characteristics

After making the connections as in Figiure. Power circuit supply is switched on ensuring the constant load resistance, and keeping the anode circuit variable rheostat at minimum position. The gate voltage is kept at minimum voltage position and supply switched on. The gate voltage is carefully adjusted to get sufficient gate current to turn on the SCR. Ensure that the SCR is conducting(check the ammeter and voltmeter status). Note down the anode current and the anode to cathode voltage. The anode current is decreased in steps by increasing the variable rheostat and note the anode currents and the voltages across the SCR in each step. Suffient current may be required to make noticeable drop across the device. Compute the ON state resistance of the SCR.

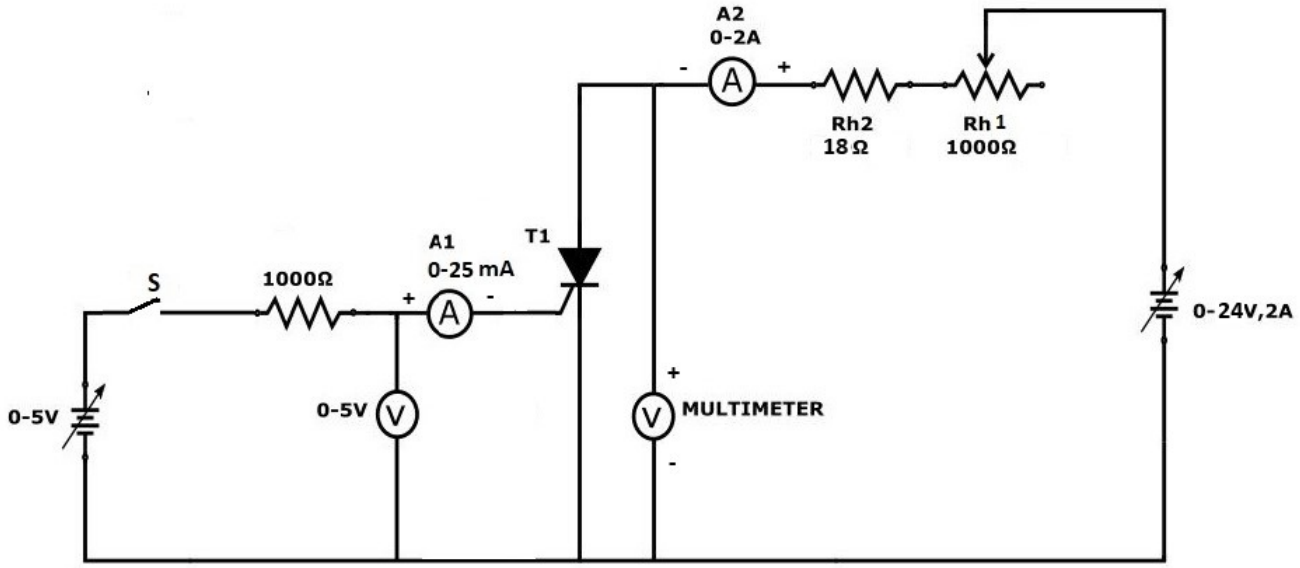


Figure 1.3: Circuit to plot forward conduction characteristics

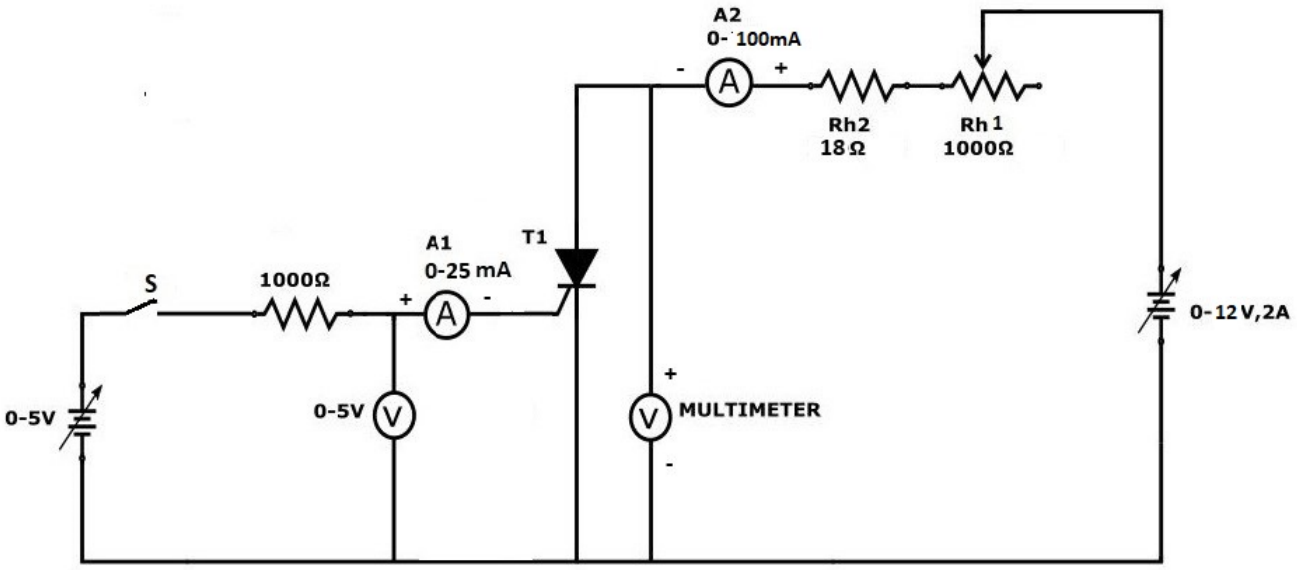


Figure 1.4: Circuit to determine latching and holding currents

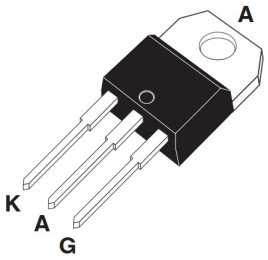


Figure 1.5: Pin details of TYN616

1.4.3 Find Latching and Holding Current

Latching current is the minimum anode current required to bring an SCR in to conduction state from blocking state (OFF state) even after the removal of the gate signal during turning on process.

Holding current is the minimum anode current required for an SCR to continue in the conduction state in the absence of gate signal. Holding current is slightly less than the latching current.

It is easy to find the holding current first than the latching current. To find the holding current, make the connections as in Fig, with a fixed load resistance and variable rheostat in minimum resistance poistion. Keep proper anode to cathode voltage in the power circuit. Then trigger SCR by applying the gate signal. After ensuring the conduction of the SCR remove the gate signal by Switching-Off the gate supply. Now gradually decrease anode current by increasing the variable rheostat. Observe the minimum value of anode current at which, the SCR suddenly falls to OFF- state from the conduction state(Observe ammeter and multimeter). If required reduce the power circuit voltage to bring the anode current to the holding current value. Also use a low range ammeter carefully to measure the Holding current if it is very low(adjust the power supply to low value if required).

Latching current is slightly more than the holding current. To find the latching current, set the power circuit voltage and resistance to keep an anode current slightly greater than holding current. Switch On the power circuit supply.Ensure that the SCR is OFF. Then apply the gate signal and remove it. Check whether the SCR has latched to conduction. If yes switch off the supply to bring the SCR to OFF state. Repeat the procedure after adjusting the circuit resistance or supply voltage to find the minimum current at which the SCR latch to conduction from OFF state, even after the removal of gate signal. Ensure that the SCR is OFF before each attempt. Record it as the Latching current.

1.5 Tabular Column

$I_{g1} =$

Sl.No.	$V_{AK}(V)$	$I_A()$

Latching Current,$I_L =$

Holding Current,I_H =

The ON state resistance of the SCR,R_{ON} =.

1.6 Result

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1.7 Simulation Result

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1.8 Inference

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1.9 Outcome

After doing this experiment, the student should be able to design experiments to obtain the static characteristics of SCR and other power devices and determine its different parameters.

1.10 Questions

1. Explain the working of SCR using two transistor analogy.
2. Can you apply a forward voltage across the thyristor soon after the reverse recovery current drops to zero? Justify your answer.
3. List the important specifications/ratings of an SCR.
4. What are the conditions for successful turn-on of an SCR?
5. What are the conditions for successful turn-off of an SCR?

Chapter 2

Single-Phase half-wave-controlled rectifier feeding R load

2.1 Objective

Aim: To set up a half-wave-controlled rectifier with line synchronized R and RC firing circuits and plot relevant waveforms such as voltage waveform across the load and thyristor, and gate voltage, for different firing angles. Find the dc output voltages, maximum and minimum firing angles. Simulate the circuit using any suitable simulation software and compare the results.

R-Triggering Circuit

2.2 Theory

The resistance trigger circuit is the simplest means of triggering a thyristor from the main supply. The circuit provides phase control essentially from 0 to 90 electrical degrees of the anode voltage.

Fig 2.1 shows the basic arrangement of triggering a thyristor using a resistance triggering circuit. The thyristor will trigger when the minimum gate current is applied and gate voltage is available. In the circuit shown the condition is reached when the instantaneous anode voltage is related as

$$e = RI_{GT} + V_{GT} + V_D$$

Where, I_{GT} -Minimum gate current in amperes to trigger the thyristor at the prevailing junction temperature and the instantaneous voltage.
V_{GT} -The minimum gate voltage to trigger the thyristor corresponding to I_{GT} and
V_D - Voltage drop of the diode D_1.

The diode D1 is provided to prevent reverse voltage from being applied to the gate during the reverse part of the anode voltage. The diode should have a peak inverse voltage rating

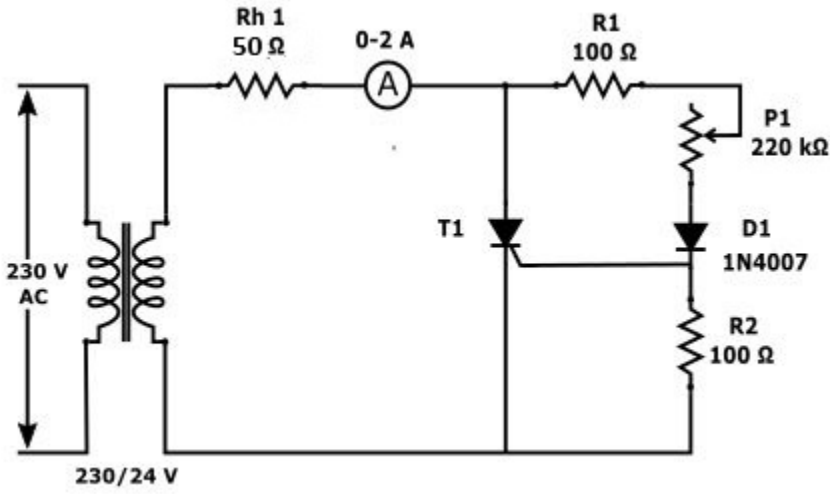


Figure 2.1: R trigger circuit

greater than the peak value of the anode voltage.

As the potentiometer P1 is varied the thyristor will trigger in accordance with the equation shown above. The circuit provides continuously variable control for the thyristor from full ON (firing angle 0°) to half ON (firing angle 90°).

2.3 Design

The design of this circuit is done by selecting 2P4M (SCR) . Some essential parameters of these devices are provided below (for more details, please refer the Data sheet)

Gate Threshold Voltage,$V_{GT} = 0.8V$
Gate Threshold Current, $I_{GT} = 200\mu A$
Max. Anode Current, $I_{Amax} = 2A$
Max. Gate Current, $I_{Gmax} = 200mA$
Max. Gate Voltage, $V_{Gmax} = 6V$

$$R_1 >= \frac{V_{max}}{I_G max} \tag{2.1}$$

For firing angle α

$$R_{max} = \frac{V_m \sin\alpha - 0.7}{200\mu A} \tag{2.2}$$

Where $R_{max} = R_1 + P_1$

$$R_2 <= \frac{V_{GM} R_1}{V_m - V_{GM}} \tag{2.3}$$

2.4 Procedure

Note down the minimum gate current and the minimum gate voltage of the thyristor from the datasheet. Calculate the value of R from the equation knowing the maximum value of the anode voltage. Set up the circuit and take down the waveform across the thyristor T1 and load (Rh1).

RC Triggering Circuit

2.5 Theory

The resistance triggering circuit provides phase-controlled delay of 0 to 90 electrical degrees only. RC triggering circuit provides control for 0 to 180 degrees of the anode voltage.

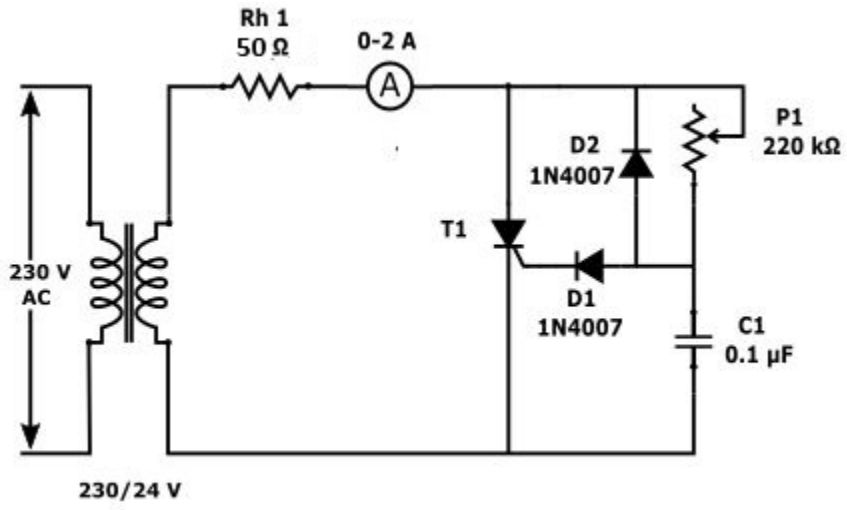


Figure 2.2: RC triggering circuit

Fig. 2.2 shows RC triggering circuit. On the positive half cycle of the thyristor anode voltage, the capacitor will charge to the triggering point of the thyristor. The time required to reach the trigger point is determined by the time constant RC and the rising anode voltage.

On the negative half cycle, the top plate of the capacitor will charge to peak value of the negative voltage cycle through D2. Thus, resetting it for the next charging cycle.

The value of RC is chosen empirically as,

$$R_{max} = \frac{V_m - 0.7}{200\mu A} \tag{2.4}$$

$$RC \approx 1.3T/2$$

where $T = 1/f$.

When $V_C = V_{GT} + V_{D1}$ the thyristor will trigger, provided gate current I_{GT} is available. Since the anode voltage across capacitor is essentially constant at the instant of triggering, the current I_{GT} must be instantaneously supplied through resistor R.

Hence, R < (e - V_{D1} - V_{GT}) / I_{Gtmin}

2.6 Procedure

Design values for R and C for the given thyristor and anode voltage. Set up the circuit and test it. Observe the voltage across thyristor, load and capacitor for different firing angles. Plot the waveforms for a particular firing angle.

2.7 Design

The design of this circuit is done by selecting 2P4M (SCR) . Some essential parameters of these devices are provided below (for more details, please refer the Data sheet)

Gate Threshold Voltage,$V_{GT} = 0.8$ V;
Gate Threshold Current, $I_{GT} = 200\mu\ A$
Max. Anode Current, $I_{Amax} = 2\ A$
Max. Gate Current, $I_{Gmax} = 200\ mA$
Max. Gate Voltage, $V_{Gmax} = 6\ V$

$$R_{max} = \frac{V_m - 0.7}{200\mu A} \tag{2.5}$$

$$RC \approx 1.3T/2$$

where $T = 1/f$.

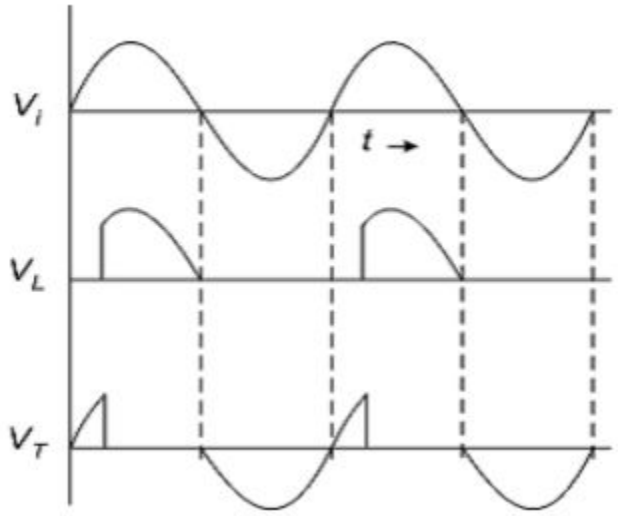


Figure 2.3: Waveform of R triggering circuit

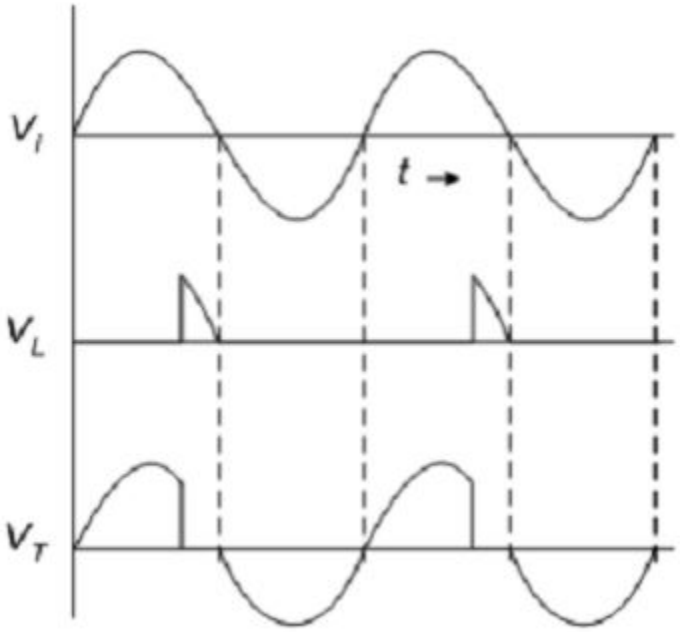


Figure 2.4: Waveform of RC triggering circuit

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2.9 Simulation Result

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2.10 Inference

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2.11 Outcome

Students should be able to design and set up the R and RC triggering circuits and analyse the output waveforms.

2.12 Questions

1. What is the order of magnitude of the on-state voltage of thyristor?
2. "A conducting thyristor can be turned off by applying a negative pulse at the gate". Is this statement true? If not, how a conducting thyristor can be turned off?
3. What are the advantages and disadvantages of simple R, RC triggering circuits?
4. What are the disadvantages of RC triggering circuit? How this is overcame in RC circuit using Diac?

5. How the power circuit can be modified to achieve full-wave controlled rectification with a single SCR and still using R or RC triggering circuit?

Chapter 3

Single Phase fully Controlled SCR bridge rectifier

3.1 Objective

Aim: To set up a line synchronized Triggering circuit for a single-phase full controlled rectifier feeding RL load and observe relevant waveforms with and without freewheeling diode. Find the output dc voltage in each case. Simulate the circuit using any suitable simulation software and compare the results.

3.2 Theory

The full controlled rectifier has four thyristors. At a time two thyristors, one from top and the other from bottom will conduct. During positive half cycles of input voltage, thyristors T1 and T3 are triggered at $\omega t = \alpha$; similarly, T2 and T4 are triggered at $\omega t = \pi + \alpha$. For a full converter the output is present during positive and negative half cycles. The average output voltage can be either positive or negative which depends on firing angle.

Figure 3.2 shows the voltage across different devices and the dc output voltage during each of these conduction modes.

Under normal operating condition of the converter the load current may or may not remain zero over some interval of the input voltage cycle. If i_0 is always greater than zero then the converter is said to be operating in the continuous conduction mode. In this mode of operation of the converter $T_1 - T_2$ and $T_3 - T_4$ conducts for alternate half cycle of the input supply. However, in the discontinuous conduction mode none of the thyristors conduct over some portion of the input cycle. The load current remains zero during that period.

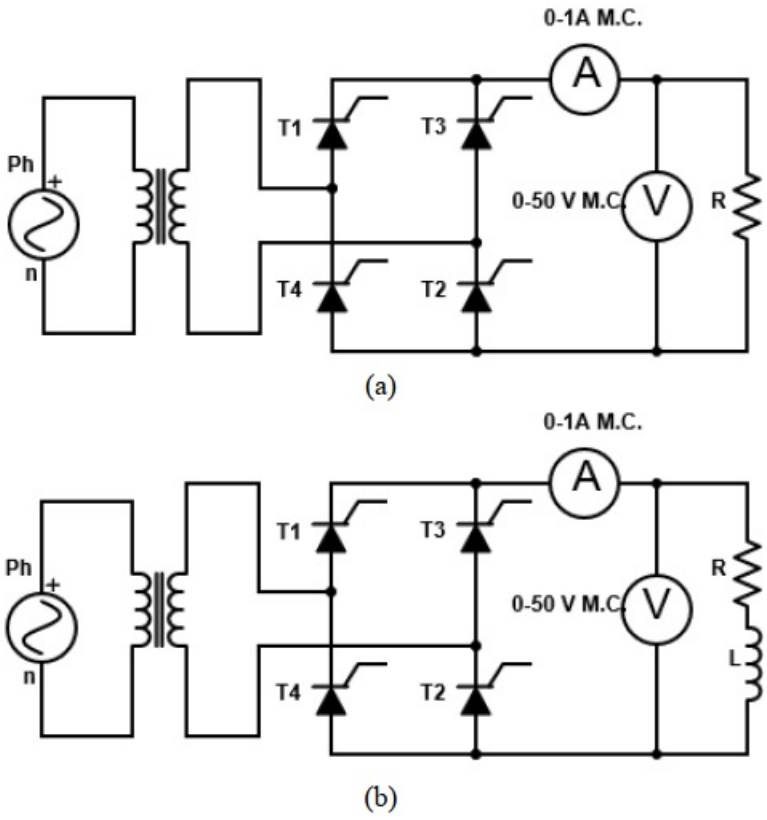


Figure 3.1: Power circuit of full converter with (a) R load and (b) R-L load.

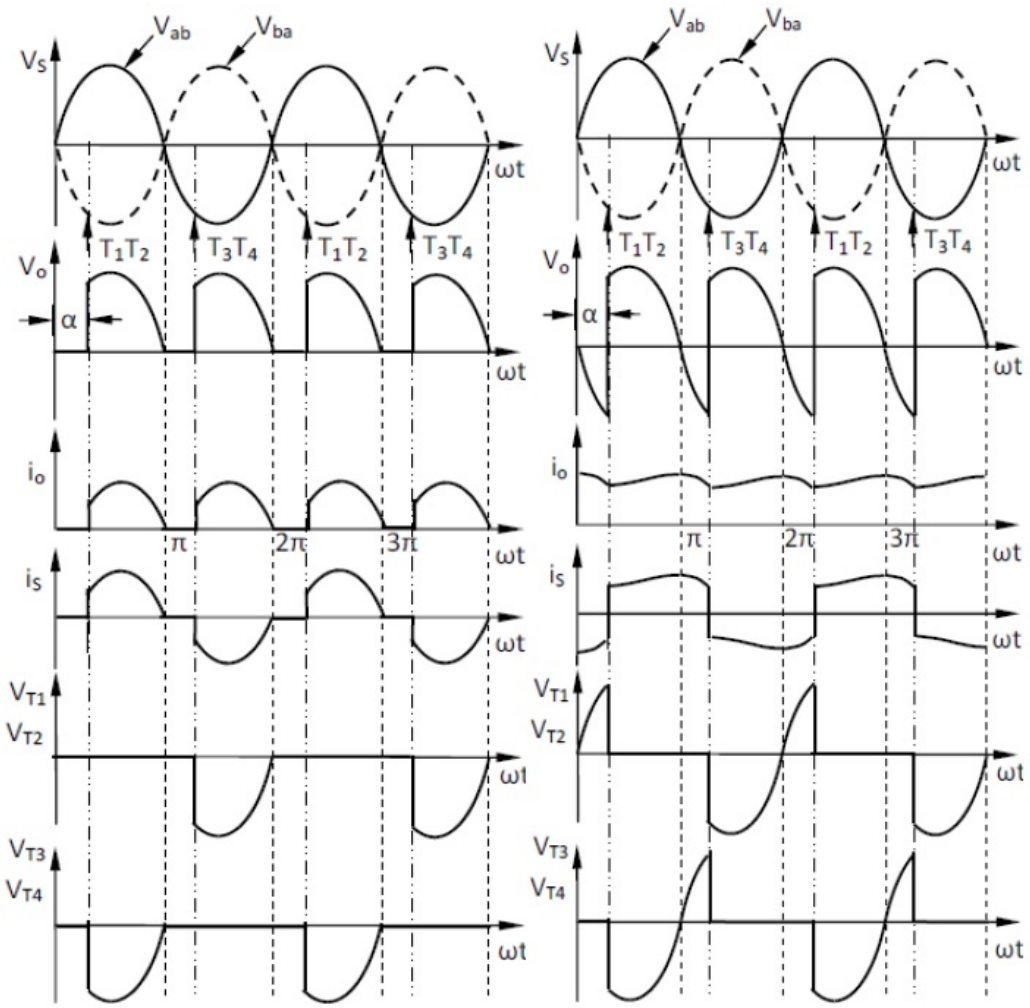


Figure 3.2: Output and SCR voltage wave forms of full bridge converter for (a) R and (b) RL loads

Sl.No.	$\alpha, Firing Angle(Degree)$	Computed output voltage(V)	Measured output voltage(V)

3.3 Procedure

Set up the circuit. Connect the power supply to the rectifier through isolation transformer to avoid any ground looping. Observe the waveforms with R load and RL load without free wheeling diode. Vary firing angle and observe the wave forms across the load and thyristors. Compare the average out put voltage obtainced by computation from wave form and by actual measurement. Repeat the experiment after connecting the free wheeling diode. Replace the load with a DC motor which can act as a RLE load or RLE load. Repeat the earlier procedure and obseve the output variation with firing angle variation.

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3.5 Simulation Result

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3.6 Inference

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3.7 Outcome

After doing this experiment, the student will be able to understand the role of free whwwling diode and control of DC out put voltage using a fully controlled rectifier.

3.8 Questions

- What is the difference in current and voltage wave forms for R and RL load with and without freewheeling diode?

Chapter 4

AC Voltage Controller Using TRIAC

4.1 Objective

Aim: To set up a single-phase AC voltage controller using TRIAC feeding a lamp load and to observe relevant waveforms such as voltage waveforms across the load and TRIAC, gate voltage for different firing angles. Find the output ac voltage for different firing angles. Simulate the circuit using any suitable simulation software and compare the results.

4.2 Theory

A Triac is a bi-directional device. The term cathode and anode are not applicable and connections are simply designed by numbers (MT_1 and MT_2). The volt ampere characteristics lies in the first and third quadrants. The device is normally triggered by a positive signal at the gate when MT_2 is positive and by a negative signal, when MT_2 is negative with respect to MT_1 (other modes of triggering are also possible). Fig. 4.1 shows the static v-i characteristics.

RC phase control circuit can be used for triggering a Triac. This circuit, however, is not stable and usually requires a large value of capacitor. A much better circuit is one that uses a Diac as trigger device.

Diac's are available with breakdown voltage in the range of 10-40 volts. Fig. 4.2 shows a phase control circuit using Triac. As the voltage across the capacitor reaches the breakdown value of Diac in either direction, a sudden discharge pulse from the capacitor triggers the Triac. The trigger current is not dependent on the resistance value as in the case of RC triggering. Therefore, a large value of resistance and smaller capacitance can be used. As the breakdown point of Diac is very stable, the trigger point also is very stable.

4.3 Design

The design of this circuit is done by selecting BT136 (TRIAC) and DB3 (DIAC). Some essential parameters of these devices are provided below (for more details, please refer the

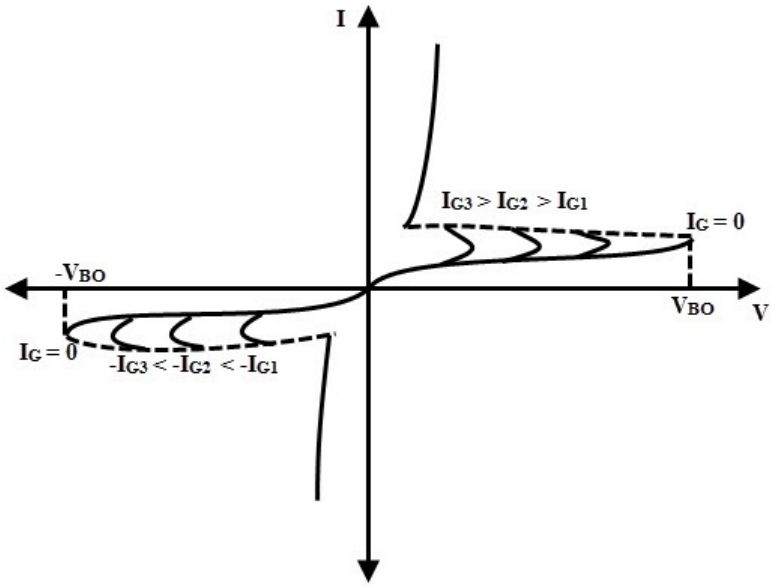


Figure 4.1: VI characteristics of TRIAC

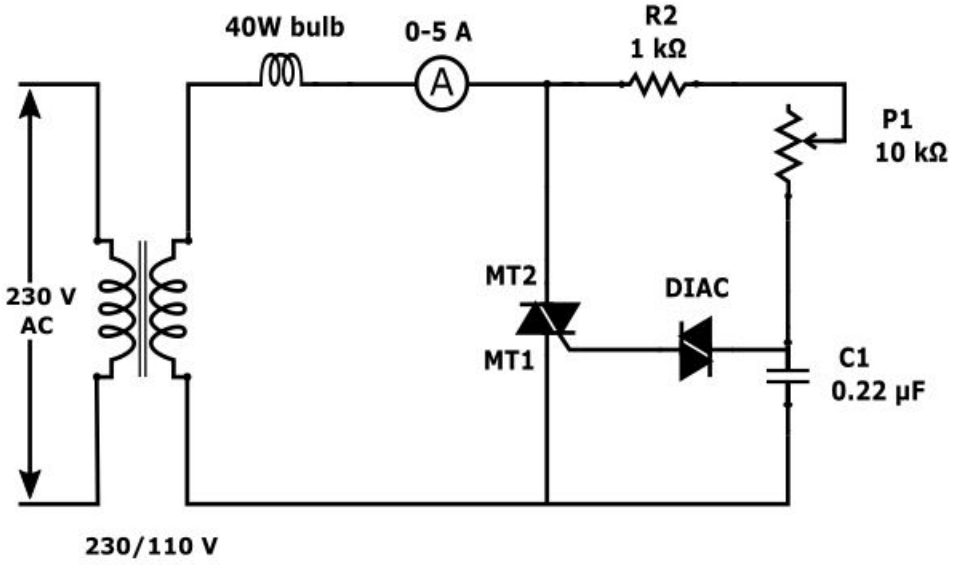


Figure 4.2: Triac firing circuit using Diac

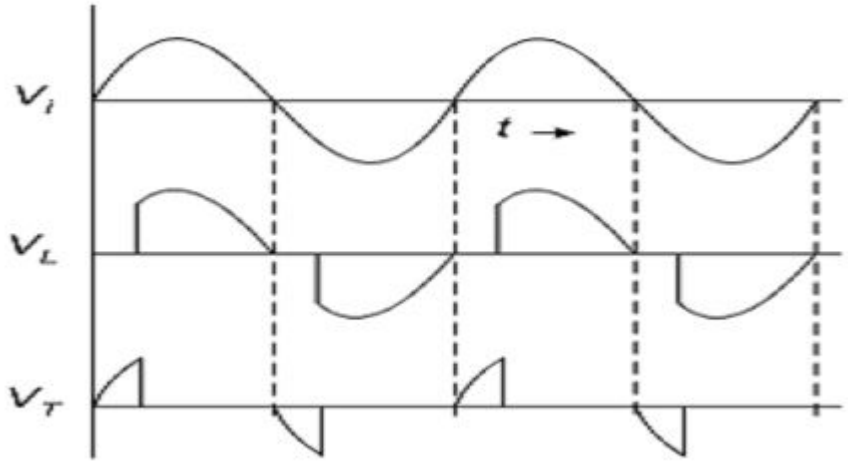


Figure 4.3: Triac circuit waveforms

Data sheet)

BT136

Gate Threshold Voltage,$V_{GT} = 0.25\ V$

Gate Threshold Current, $I_{GT} = 35\ mA$

Max. Gate Current, $I_{GM} = 2\ A$

DB3

Threshold Voltage, $V_T = 32\ V$

$$R = R_2 + P_1$$

$$R <= \frac{V_S - V_{GT} - V_T}{I_G T} \tag{4.1}$$

Peak value of Input Voltage Waveform , $V_S = 110\sqrt{2}$

$$RC >= \frac{1.3T}{2} \tag{4.2}$$

$$R_2 = \frac{V_S}{I_{GM}} \tag{4.3}$$

4.4 Procedure

Setup the circuit as shown in Fig.4.2. Adjust the fire angle using the potentiometer and trace the waveforms of load voltage, voltage across Triac etc. Care has to be taken while

using two-channel oscilloscopes, as the channels are not isolated electrically. Hence, when two waveforms are to be simultaneously seen on the scope, the probe grounds shall be connected only to one point. The design of the RC network can be done with the same empirical relation used for RC triggering circuit. It is to be noted that the capacitor charges to both polarities of voltage, hence it should not be an electrolytic capacitor.

4.5 Result

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4.6 Simulation Result

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4.7 Inference

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4.8 Outcome

Student should be able to understand the working of TRIAC and its output waveforms.

4.9 Questions

1. Compare the performance of a triac with that of two SCR connected in anti-parallel.
2. Is it possible to connect a FWD at the output of a triac controller?
3. Experimentally find the rms value of voltages across the load for a firing angle of 30 degree and 60 degree.